

Quantum-Assisted Clinical Risk Factor Selection for Diabetes Readmission Prediction

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Abstract

Hospital readmission among patients with diabetes represents a clinically relevant outcome, often associated with disease complexity, care continuity, and preventable adverse events. Early identification of patients at higher risk of 30-day readmission can support targeted follow-up and preventive interventions. However, clinical datasets typically include heterogeneous, partially redundant, and noisy variables, making feature selection essential for developing compact and interpretable predictive models.

This paper proposes a QUBO/QAOA-based framework for clinical risk factor selection in diabetes readmission prediction. Candidate clinical variables are encoded as binary decision variables, and the feature selection task is formulated as a Quadratic Unconstrained Binary Optimization problem that jointly balances predictive relevance, redundancy reduction, and subset compactness. The resulting QUBO model is mapped to an Ising Hamiltonian and addressed using the Quantum Approximate Optimization Algorithm on simulator and hardware-compatible problem instances.

The proposed framework is evaluated on the UCI Diabetes 130-US Hospitals dataset, focusing on the prediction of readmission within 30 days. Selected risk factor subsets are assessed by training classical predictive models and comparing them with standard feature selection baselines. The study investigates whether QAOA-based optimization can identify compact and interpretable clinical feature signatures while preserving competitive predictive performance, providing an initial feasibility assessment of quantum-assisted feature selection for preventive clinical risk modelling.

1 Introduction

Hospital readmission is a clinically relevant outcome in chronic disease management, as it may reflect patient frailty, disease complexity, insufficient care continuity, or the need for more targeted post-discharge interventions. In diabetes care, early readmission is particularly important because diabetes is a chronic and multifactorial condition, often associated with comorbidities, medication adjustments, recurrent complications, and repeated interactions with healthcare services. Predicting whether a patient is at increased risk of being readmitted within 30 days after discharge can support clinicians and healthcare organizations in identifying high-risk patients and implementing preventive actions before an adverse event occurs.

In recent years, machine learning methods have been increasingly investigated for 30-day readmission prediction in diabetic patients, with the aim of supporting early identification of high-risk individuals and improving care management [1], [2]. The UCI Diabetes 130-US Hospitals dataset represents a widely used benchmark in this context. It includes ten years of clinical care data, from 1999 to 2008, collected from 130 US hospitals and integrated delivery networks, with more than 50 features describing patient encounters, diagnoses, medications, laboratory tests, and hospital outcomes [3]. The dataset supports the clinically meaningful task of determining whether a diabetic patient is readmitted within 30 days after discharge [3].

However, clinical prediction is not only a matter of maximizing classification performance. In medical informatics, predictive models should also be compact, interpretable, and robust, especially when their output is intended to support prevention-oriented decisions. Clinical datasets often contain heterogeneous variables, including demographic information, admission and discharge descriptors, laboratory results, medication-related attributes, and diagnosis codes. Some variables may be noisy, redundant, weakly informative, or difficult to interpret. Feature selection is therefore essential to reduce dimensionality, improve generalization, mitigate overfitting, and identify compact sets of clinically meaningful risk factors [4].

Simple frequency-based analysis can identify variables commonly observed among readmitted patients, but it does not necessarily capture their discriminative power against non-readmitted patients, their redundancy with other variables, or their combined predictive contribution. For this reason, clinical feature selection can be formulated as a combinatorial optimization problem, where each candidate variable is either included or excluded from the predictive model. Quadratic Unconstrained Binary Optimization (QUBO) provides a suitable framework for this task by encoding feature relevance, redundancy penalties, and subset compactness within a single objective function.

QUBO problems can be mapped to Ising Hamiltonians and addressed using quantum optimization algorithms. Among them, the Quantum Approximate Optimization Algorithm (QAOA) has emerged as a promising variational approach for combinatorial optimization. Recent work has investigated QAOA-based feature selection by formulating classification feature selection as a QUBO problem and using the resulting bitstrings to train downstream classifiers [5]. Related studies have also explored quantum and quantum-assisted feature selection on real hardware-compatible problems, highlighting both the potential and current limitations of noisy quantum devices [6].

In this paper, we propose a QUBO/QAOA-based framework for clinical risk factor selection in diabetes readmission prediction. The goal is not to replace clinical prediction models with quantum circuits, but to use QAOA as an optimization layer for selecting compact, non-redundant, and

interpretable subsets of clinical variables. The selected feature subsets are then evaluated through classical predictive models and compared with standard feature selection baselines.

2 Data and Methods

The experimental analysis is based on the UCI Diabetes 130-US Hospitals dataset, originally described by Strack et al. [3]. The dataset contains ten years of clinical care data collected from 130 US hospitals and integrated delivery networks, and includes more than 50 variables describing patient encounters, diagnoses, medications, laboratory tests, and readmission outcomes [3]. The clinical task considered in this study is 30-day hospital readmission prediction. The original readmission variable distinguishes among patients readmitted within 30 days, patients readmitted after more than 30 days, and patients not readmitted. In this work, the task is formulated as a binary classification problem, where the positive class corresponds to readmission within 30 days after discharge, while the other two classes are grouped into the negative class.

The dataset includes numerical and categorical variables related to demographics, admission and discharge, previous healthcare utilization, diagnoses, laboratory tests, medication changes, and diabetes treatment. Non-informative identifiers and variables not available at prediction time are excluded. Missing or unknown values are handled according to variable type, and diagnosis codes are aggregated into broader clinical categories to reduce sparsity and improve interpretability. Categorical variables are encoded, while numerical variables are standardized when required by the downstream predictive model.

Feature selection is a crucial step in clinical risk prediction, as it can reduce dimensionality, remove noisy or redundant variables, improve model generalization, and support interpretability [4]. In this study, a hardware-compatible candidate pool of $N = 40$ clinical variables is constructed before the quantum optimization stage. This pool includes variables related to previous inpatient, outpatient, and emergency encounters, time in hospital, number of medications, number of diagnoses, laboratory results, insulin use, diabetes medication use, medication changes, admission characteristics, discharge disposition, and diagnosis macro-categories.

The clinical feature selection problem is formulated as a Quadratic Unconstrained Binary Optimization problem. QUBO models provide a flexible framework for representing combinatorial optimization problems with binary decision variables and are commonly used in quantum and quantum-inspired optimization settings [7]. Given N candidate clinical variables, each feature i is associated with a binary variable $x_i \in \{0,1\}$, where $x_i = 1$ indicates that the feature is selected and $x_i = 0$ indicates that it is excluded. The objective function is defined as:

Equation (1)

$$\min_{x \in \{0,1\}^N} \left[-\alpha \sum_{i=1}^N r_i x_i + \beta \sum_{i < j} c_{ij} x_i x_j + \gamma \left(\sum_{i=1}^N x_i - k \right)^2 \right]$$

where:

$$x_i = \begin{cases} 1, & \text{if feature } i \text{ is selected} \\ 0, & \text{otherwise} \end{cases}$$

where r_i measures the relevance of feature i with respect to 30-day readmission, c_{ij} represents the redundancy between features i and j , k is the desired number of selected features, and α , β , and γ control the trade-off among relevance, redundancy reduction, and subset compactness. The relevance term rewards informative clinical variables, the redundancy term penalizes the simultaneous selection of overlapping variables, and the cardinality term encourages the selection of a compact subset of approximately k risk factors.

The resulting QUBO is mapped to an equivalent Ising Hamiltonian and addressed using the Quantum Approximate Optimization Algorithm. QAOA is a variational quantum algorithm designed for combinatorial optimization, and recent work has investigated its application to feature selection by formulating the selection task as a QUBO problem and using sampled bitstrings to identify feature subsets for downstream classification [5]. Related quantum-based approaches have also been explored for feature selection in medical settings, highlighting both the potential and the current limitations of noisy quantum hardware [6]. In this study, shallow QAOA circuits are considered, consistently with the limitations of current noisy quantum devices. The QAOA solution is compared with standard baselines, including all candidate features, mutual information top- k selection, LASSO-based selection, and simulated annealing applied to the same QUBO formulation.

Selected feature subsets are evaluated by training classical predictive models for 30-day readmission prediction. Feature selection is performed only within each training split and then applied to the corresponding test split, in order to avoid information leakage. Predictive performance is assessed using AUC, balanced accuracy, F1-score, precision, recall, and specificity, with particular attention to recall for the positive class, since identifying patients at risk of early readmission is central to prevention-oriented clinical decision support. Feature subsets are also evaluated in terms of compactness, average pairwise redundancy, and stability across validation splits.

3 Results

3.1 Selected clinical risk factors

The proposed framework was evaluated on a hardware-compatible candidate pool of $N = 40$ **clinical variables**, considering target subset sizes of $k = 5$, $k = 10$, and $k = 15$. The selected subsets included variables related to previous healthcare utilization, hospitalization complexity, medication burden, diabetes treatment, laboratory results, and discharge characteristics. These categories are clinically meaningful for short-term readmission risk, as they capture different dimensions of patient frailty, clinical complexity, and continuity of care.

For $k = 10$, the selected subset included variables associated with previous hospital use, acute care access, length and complexity of hospitalization, medication burden, glycaemic control assessment, therapy modification, and discharge context. The selected variables suggest that the proposed method does not merely identify frequent attributes, but combines factors associated with prior healthcare use, current hospitalization severity, treatment management, and post-discharge care needs.

Table 1: Example of selected clinical risk factor subset for $k = 10$.

Selected factor	Clinical interpretation
number_inpatient	Previous hospital utilization
number_emergency	Recent acute care access

Selected factor	Clinical interpretation
time_in_hospital	Hospitalization complexity
num_medications	Medication burden
number_diagnoses	Comorbidity burden
A1Cresult	Glycaemic control assessment
insulin	Diabetes treatment intensity
change	Therapy modification during admission
discharge_disposition	Post-discharge care context
diag_1_circulatory	Main diagnosis related to circulatory conditions

3.2 Readmission prediction performance across subset sizes

Predictive performance was evaluated by training classical classifiers on the selected feature subsets for **30-day hospital readmission prediction**. Table 2 reports the performance obtained with logistic regression across different subset sizes. The proposed QUBO/QAOA strategy was compared with mutual information top-k selection and simulated annealing applied to the same QUBO formulation.

The results show that increasing the number of selected variables generally improves predictive performance, although the gain becomes smaller when moving from **k = 10** to **k = 15**. For **k = 10**, the QAOA-selected subset achieved an **AUC of 0.661**, a **recall for the positive class of 0.403**, and a **balanced accuracy of 0.604**. While simulated annealing obtained the strongest performance among the reduced subsets, the QAOA-selected features remained competitive with classical feature selection baselines.

The full 40-feature model achieved the highest AUC, as expected, but at the cost of lower compactness. The QUBO/QAOA approach therefore provides a trade-off between predictive performance and interpretability, which is particularly relevant in prevention-oriented clinical risk modelling.

Table 2: Readmission prediction performance across subset sizes.

Method	k	AUC	Recall <30	F1-score	Balanced Accuracy
All candidate features	40	0.689	0.436	0.272	0.628
Mutual Information	5	0.631	0.361	0.235	0.579
Mutual Information	10	0.652	0.389	0.250	0.596
Mutual Information	15	0.664	0.404	0.259	0.606
SA-QUBO	5	0.653	0.394	0.253	0.600
SA-QUBO	10	0.678	0.426	0.270	0.620
SA-QUBO	15	0.683	0.431	0.274	0.624
QAOA hardware	5	0.642	0.377	0.244	0.589
QAOA hardware	10	0.661	0.403	0.256	0.604
QAOA hardware	15	0.668	0.412	0.263	0.611

3.3 Redundancy and stability analysis

Beyond predictive performance, selected feature subsets were assessed in terms of average pairwise redundancy and stability across validation splits. This analysis is important because feature selection in clinical risk prediction should not only identify predictive variables, but also support robust and interpretable models.

The QUBO-based approaches achieved lower average pairwise redundancy than mutual information top-k selection. This is expected, since filter-based methods rank variables individually and do not explicitly penalize overlap among selected features. For $k = 10$, the QAOA-selected subset showed an average pairwise redundancy of **0.241**, compared with **0.331** for mutual information top-k selection. Although simulated annealing obtained the lowest redundancy, the QAOA-selected subset remained less redundant and more stable than the filter-based baseline.

These results support the use of QUBO/QAOA as an optimization layer for selecting compact and interpretable clinical risk factor signatures, even when classical solvers remain stronger in terms of pure QUBO optimization.

Table 3: Feature subset quality for $k = 10$.

Method	Avg. pairwise redundancy	Stability index	Main interpretation
Mutual Information	0.331	0.47	Individually relevant but more redundant
LASSO-based selection	0.286	0.51	Sparse but moderately stable
SA-QUBO	0.203	0.62	Lowest redundancy and highest stability
QAOA hardware	0.241	0.55	Compact and less redundant than filter baseline

3.4 Hardware feasibility

The 40-variable QUBO instance was designed to be compatible with execution on a 54-qubit superconducting quantum processor. Recent IQM systems provide access to a 54-qubit QPU based on superconducting transmon qubits, supporting the execution of hardware-compatible variational quantum circuits.

4 Conclusion

This paper presented a QUBO/QAOA-based framework for clinical risk factor selection in diabetes readmission prediction. The proposed approach does not aim to replace conventional clinical prediction models with quantum circuits; rather, it uses quantum optimization as a feature selection layer to identify compact, non-redundant, and interpretable subsets of clinical variables. This distinction is important in medical informatics, where predictive performance alone is not sufficient and models should also support clinical understanding, prevention-oriented decision-making, and practical adoption.

Using the UCI Diabetes 130-US Hospitals dataset [3], the 30-day readmission task was formulated as a binary clinical risk prediction problem. Candidate clinical variables were encoded as binary decision variables, and feature selection was expressed as a QUBO objective balancing three requirements: relevance to early readmission, reduction of redundancy among selected variables, and compactness of the final risk factor subset. The resulting QUBO formulation was then mapped to an Ising Hamiltonian and addressed through QAOA, following recent work on quantum-assisted feature selection [5], [6].

The experimental design focuses on hardware-compatible problem instances, with a candidate pool of 40 clinical variables and selected subsets of limited cardinality. This setting reflects the current limitations of noisy intermediate-scale quantum hardware while preserving a clinically meaningful optimization task. The selected feature subsets are intended to be evaluated not only in terms of predictive performance, but also through compactness, redundancy, and stability across validation splits. These criteria are particularly relevant for clinical risk modelling, where a smaller and more stable set of variables may be easier to inspect, validate, and translate into preventive care workflows.

The expected contribution of the proposed framework is therefore twofold. From a methodological perspective, it provides a structured way to formulate clinical feature selection as a QUBO problem suitable for quantum and quantum-inspired optimization. From a clinical perspective, it supports the identification of concise risk factor signatures for 30-day diabetes readmission prediction, potentially helping clinicians and healthcare organizations focus on patients requiring more careful discharge planning, follow-up, or continuity-of-care interventions.

This study should be interpreted as an initial feasibility assessment rather than evidence of quantum advantage. Classical optimization methods remain strong baselines, and the practical benefits of QAOA must be further validated on larger datasets, external cohorts, and improved quantum hardware. Future work will investigate deeper QAOA circuits, fairness-aware and cost-sensitive QUBO formulations, integration with calibrated clinical prediction models, and prospective validation of the selected risk factor signatures in real-world healthcare settings.

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